MIL-O-22039C
15 November 1978
SUPERSEDING
MIL-O-22039B
25 September 1970

MILITARY SPECIFICATION

OPAQUES, PHOTOGRAPHIC FILM-PLATE RETOUCHING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers opaques in paste form.

1.2 Classification. Opaques shall be furnished in the type, pigments, form, and packaged units shown in Table I, as specified (see 6.2):

TABLE I. Opaques.

Type vehicle	Pigment	Form	Packaged units
I (Water)	Graphite	Paste	2-ounce (59.1 milliliters (ml)) jar
	Red rouge	Paste	2-1/2-ounce (74 ml) jar
	Red rouge	Paste	8-ounce (236.6 ml) jar

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6750

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-P-1892	- Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

STANDARDS

FEDERAL

FED. STD. No. 1	- Standard for Laboratory Atmospheric Conditions for Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Opaques shall be colloidal dispersion of opaque pigment. The vehicle used for thinning Type I shall be water. The pigments shall be as specified in 1.2.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3 Storage stability. When tested as specified in 4.4.2.2, the opaques shall not be separated, crystallized, or lumpy, and containers shall not be cracked, chipped, broken, split, or unable to be opened.

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3.4 Working properties. Opaques shall brush freely and smoothly with no tendency to pull.

3.5 Drying. When allowed to dry for 20 minutes, opaques shall not be sticky, shall leave no mark when touched by the finger, and shall not wipe off on a soft dry cloth.

3.6 Opaquing. When allowed to dry for 3 hours, opaques shall be uniformly opaque with a density more than 3.0 and free from light, streaks, spots, or pin holes.

3.7 Heat resistance. Opaques shall not cause cockling or buckling and shall be free from spots and pin holes when placed on photographic film and subjected to a temperature of 125° F (51.6° C), plus or minus 3° F (1.7° C).

3.8 Removal. Opaques shall be completely removable from glass plates and film after drying for 18 hours.

3.9 Evaporation. After being left uncovered for 5 hours, opaques shall be restorable to a smooth consistency.

3.10 Pigment settlement. When allowed to stand undisturbed for 48 hours opaques shall show no separation of liquid.

3.11 Paste opaque thinning. Opaques shall thin to a uniform consistency.

3.12 Identification marking. Containers shall be identified in accordance with MIL-STD-130.

3.13 Instructions. Each container shall be permanently and legibly marked with instructions for use.

3.14 Workmanship. Opaques shall be free of foreign materials and any defects and imperfections which might impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of all opaque manufactured as one batch and offered for delivery at one time.

4.3.2 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.4 Test. Samples selected in accordance with 4.3.2 shall be tested as specified in 4.4.2. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The opaque and container shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Container not as specified.
103. Instructions not included.
104. Identification marking, missing, incomplete, or illegible.
105. Workmanship not as specified.

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4.4.2 Tests.

4.4.2.1 Test condition. Prior to application, opaque shall be thinned in accordance with the contractor's instructions. A 0.5-inch (in.) (12.7 millimeter (mm)) size artist's brush shall be used to apply opaques. Glass plates shall be thoroughly cleaned with acetic acid and water, and thoroughly dried before use. The atmospheric condition of the room in which tests are conducted shall be in accordance with FED. STD. No. 1.

4.4.2.2 Storage stability. The opaque in containers shall be placed in a temperature test chamber and the temperature shall be raised to 155° F (68.3° C) for 4 hours. The temperature in the chamber shall then be lowered to 36° F (2.2° C) for 3 days. The temperature in the chamber shall then be raised to approximately 70° F (21° C) and the opaque and the containers shall be examined for damage. Nonconformance to 3.3 shall constitute failure of this test.

4.4.2.3 Working properties. Glass plates prepared as specified in 4.4.2.1 shall be used. Heavy opaqued lines and spots shall be placed on the glass plates. Any evidence of the opaques not brushing freely and smoothly shall constitute failure of this test. The opaqued plate shall be retained for the test specified in 4.4.2.7.

4.4.2.4 Drying. A glass plate prepared as specified in 4.4.2.1 shall be used. A 7 cm (2.75 in.) square shall be marked on a glass plate with a wax pencil. Not more than 2 ml (0.07 oz) original volume of opaque shall be brushed evenly on the square. The sample shall be dried for 20 minutes. Any evidence of the opaque being sticky when touched by a finger or wiping off on a soft dry white cloth shall constitute failure of this test.

4.4.2.5 Opaquing. An opaqued glass plate prepared as specified in 4.4.2.4 shall be allowed to dry for 3 hours, and the density shall be measured with a densitometer. A reading of less than 3 on the densitometer or any light streaks, spots, or pin holes shall constitute failure of this test.

4.4.2.6 Heat resistance. A thin coat of opaque shall be brushed on a transparent photographic film. The opaqued film shall be subjected to a temperature of 125° F (51.6° C), plus or minus 3° F (1.7° C), for 10 minutes and examined in front of a strong light. Any evidence of

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cockling, buckling, spots, or pin holes shall constitute failure of this test. The opaqued film shall be retained for the test specified in 4.4.2.7.

4.4.2.7 Removal. The opaque applied as specified in 4.4.2.3 and 4.4.2.6 shall be allowed to dry for not less than 18 hours and then rubbed lightly with a cloth wetted with thinner. Inability of the thinner to remove the opaque from the glass plate or film shall constitute failure of this test.

4.4.2.8 Evaporation. A 100 by 10 mm (39.4 by 0.4 in.) deep Petri dish or equivalent shall be filled to a depth of 7 to 8 mm (0.27 to 0.31 in.) with opaque and left uncovered for 5 hours. Thinner shall be added slowly with constant stirring to restore the original volume. Inability of the opaque to be restored to a smooth consistency shall constitute failure of this test.

4.4.2.9 Pigment settlement. A closed original container of opaque shall be allowed to stand undisturbed for 48 hours. The container shall then be opened and examined for separation. Evidence of any separation shall constitute failure of this test.

4.4.2.10 Opaque thinning. A 20 ml (0.68 oz) sample of opaque shall be placed in a small beaker, and not less than 5 ml (0.17 oz) of thinner shall be added slowly with constant stirring. Inability of the opaque to be thinned to a uniform consistency shall constitute failure of this test.

4.5 Inspection of packaging. The preservation, packing, and marking of the opaques for Level A or Level B shall be examined for compliance with the quality assurance provisions of PPP-P-1892. The preservation, packing, and marking for Commercial shall be examined for compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.1.1 Level A. Opaques contained in jars, as specified in Table I, shall be intermediate packaged in accordance with the Level A packaging requirements for PPP-P-1892, as specified for containers.

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5.1.2 Level B. Opaques contained in jars, as specified in Table I, shall be intermediate packaged in accordance with the Level B packaging requirements of PPP-P-1892, as specified for containers.

5.1.3 Commercial. Preservation of the opaques contained in jars, as specified in Table I, shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Packing shall be in accordance with the Level A requirements of PPP-P-1892, except that wood-cleated fiberboard, wood-cleated paper, overlaid veneer, and fiberboard boxes are not acceptable for Level A.

5.2.2 Level B. Packing shall be in accordance with the Level B requirements of PPP-P-1892.

5.2.3 Commercial. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with PPP-P-1892.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Opaques are intended for use in retouching photographic films or plates.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, pigment, form, and packaged units required (see 1.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).

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6.3 Classification change. Type II opaques have been discontinued since they are no longer required.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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Navy - AS
Air Force - 99

Review activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-O-22039C Opaques, Photographic Film-Plate Retouching

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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